Xiao Fu

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Education

Zhejiang University Hangzhou, China

Bachelor in Information Engineering | College of Information Science & Electronic Engineering *Sept. 2018 – Jul. 2022*

- **GPA:** 3.96/4.00, 90.54/100, **Ranking:** 4/145
- **National Scholarship awardee, Twice**
- **Chu Kochen Honors College:** Intensive Training Program of Innovation and Entrepreneurship (ITP) (Honored minor for the selected top 40 students with innovation capability from 6,000 freshmen)
- **Courses:** Computer Vision(96),Digital Signal Processing(95),Matrix Theory(94),Stochastic Process(96),Probability and Mathematical Statistics(95),Numerical Analysis(92),Signals and Systems(98),Calculus(A)(90)
- **Language Proficiency:** TOEFL(iBT): 106(R:29,L:27,S:24,W:26)

Publications

∗ indicates equal contribution

- [2] **Xiao Fu***, Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao, "Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation," Under Review of *European Conference on Computer Vision (**ECCV**)* 2022
- [1] Guangyi Zhang*, **Xiao Fu***, Qiyu Hu, Yunlong Cai, Guanding Yu, "Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network," *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 **[Paper][Code][Slides]**

Research Experience

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer Max Planck Institute for Intelligent Systems

Supervisor: Prof. Andreas Geiger and Prof. Yiyi Liao *Sept. 2021 – Mar. 2022*

- Propose Panoptic NeRF, which aims for obtaining per-pixel 2D semantic and instance labels from easy-to-obtain coarse 3D bounding primitives. Our method utilizes NeRF as a differentiable tool to unify coarse 3D annotations and 2D semantic cues transferred from existing datasets. This combination allows for improved geometry guided by semantic information which enables rendering of accurate semantic maps across multiple view. By inferring in 3D space and rendering to 2D labels, our 2D semantic and instance labels are multi-view consistent by design

Faster Solution of Optimization Algorithm with DLDUNN Zhejiang University

Supervisor: Prof. Guanding Yu *Jun. 2020 – Feb. 2021*

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network(DLDUNN) which unfolds the iterative Penality Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award(Team leader) - Student Research Training Program(Province-level)**

Honors and Scholarships [Certificate & Report]

- National Scholarship **(Highest Honor for Chinese Undergraduates, Twice)** *2020,2021*
- ISEE Honor Student Award **(Top 5 students among ISEE 314 undergraduates)** *2021*
- ISEE Yuelun Alumni Scholarship **(Top 10 teams out of ISEE undergraduates and graduates)** *2020,2021*
- First-Class Scholarship for Academic Excellence of Zhejiang Univeristy**(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 3%)** *2019*
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 7% out of 26,112 teams)** *2021*
- Gold Prize – 7^{th} International "Internet+" Innovation and Entrepreneurship Contest **(Team Leader)** *2021*
- Gold Prize – 12^{th} "Challenge Cup" National College Student Business Plan Competition *2020*

- Grand Prize – "Yongdian Cup" Innovation and Entrepreneurship Contest **(Team Leader)** *2019*
- Grand Prize – "Midea" Global Technology Innovation Contest *2020*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*
- Zhejiang University Five-Star Volunteer *2021*

浙江大学 信息与电子工程学院
College of Information Science & Electronic Engineering, Zhejiang University

CERTIFICATE

This is to certify that Mr Xiao Fu ranks the 4th among the 145 students with the specialty of Information Engineering.

College of Information Science & Electronic Engineering

Zhejiang University

28/09, 2021

浙江大学 信息与电子工程学院

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Zhejiang University

Student's Academic Records

Registration No: 20195473

Name: FU Xiao	College/Dept.: College of Information Science and Electronic Engineering	Speciality: Information Engineering	Student ID: 3180101153
Sex: Male	Birthday: 07/19/2000 · Birth Place: Sichuan	Entrance Date: 09/01/2018 · Graduation Date: 06/30/2022	Years of Program: 4Years

Academic Year 2018-2019		
Courses(1st Term)	***Cr**	***Sc**
Introduction to Mechanics	1.5	A+
The Concise History of Western Music	1.5	93
Military Training	2.0	89
Introduction to Modern Bionics	1.5	89
Mental Education and Foundation of Law	3.0	90
College English Band III	3.0	84
Calculus（A）Ⅰ	5.0	87
Wireless Network Application	1.5	99
Line Dance	1.0	92
Linear Algebra（A）	3.5	88
Introduction to Information Science and Electronic Engineering	2.0	89
Fundamentals of C Programming	3.0	88
Courses(2nd Term)	***Cr**	***Sc**
Introduction to Engineering Ethics(A)	1.5	97
Ordinary Differential Equations	1.0	86
Situation and Policy Ⅰ	1.0	85
College English Ⅳ	3.0	88
Calculus（A）Ⅱ	5.0	90
Study on The Book of Master Zhuang	3.0	95
Lectures on Programming	2.0	87
University Physics (A) Ⅰ	4.0	88
Cross-Country Orienteering (Basic Level)	1.0	95
Modern Chinese History	3.0	90

Electronic Engineering Training（A）	1.5	90
Engineering Graphics	2.5	70
Academic Year 2019-2020		
Courses(1st Term)	***Cr**	***Sc**
Complex Variable Functions & Integral Transformation	1.5	86
Software Technology	2.0	88
Understanding Internship	2.0	99
Electronic Circuit Design Experiments I	0.5	83
University Physics (A) Ⅱ	4.0	89
Military Theory	1.5	97
Basics of Electronics and Circuits	5.0	89
Taekwondo Training	1.0	94
Numerical Analysis	2.0	92
Engineering Training	1.5	85
College Physics Experiment	1.5	88
Probability and Mathematical Statistics	2.5	95
Courses(2nd Term)	***Cr**	***Sc**
Team Communication & Leadership	2.0	85
Optimization Algorithms	3.0	quit
Introduction to the Principle of Marxism	3.0	95
Signals and Systems	4.0	98
Electromagnetic Fields & Waves	4.0	95
Swimming (Basic Level)	1.0	98
Electronic Circuit Design Experiments II	1.0	94
Experiment of Digital System Design	1.0	96

Economics	2.0	95
Stochastic Process	1.5	96
Digital System Design	4.0	91
Discrete Mathematics	2.5	92
Academic Year 2020-2021		
Courses(1st Term)	***Cr**	***Sc**
Design Experiment of Intelligent Mobile System	2.0	100
Matrix Theory	2.0	94
Career Planning B	1.5	96
English ProficiencyTest○	1.0	P
Business Model	1.0	87
General Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	2.0	84
Intro.to Mao Thought & Theoretical System of China Socialism	5.0	90
Digital Signal Processing	3.0	95
Data Analysis and Algorithm Design	3.0	89
Information, Control & Computing	3.0	97
Classroom Ⅳ: Exchange Programs in Hong Kong, Macao, Taiwan and foreign countries○	2.0	P
Courses(2nd Term)	***Cr**	***Sc**
Principles of Embedded Computing System Design	2.0	93
Scientific Research Training Program II	2.0	quit
Photoelectric information processing comprehensive experiment	2.0	90
Lab. of Principles of Communication	1.0	93
Artificial Intelligence	3.0	91
Digital Image Processing	3.0	82
Network and Communication Security	2.0	85

IT Engineering Ethics and Project Management	1.5	92
Introduction to Data Mining	2.0	84
Principles of Communications	3.0	89
Academic Year 2021-2022		
Courses(1st Term)	***Cr**	***Sc**
Project Internship	2.0	A
iBT-Reading	1.5	93
Computer Vision	2.0	96
Physical test○	1.0	P
Classroom Ⅱ On-campus Practice Activities○	4.0	A
Classroom Ⅲ Out-of-campus (Domestic) Practice Activities○	2.0	A

浙江大学 成绩考核专用章 (1)

Overall GPA:3.96/4.0(90.54/100)

Degree Granted:	Credits Required for Graduation: 157+8+6	Credits Obtained: 165.5

Three grade systems are used simultaneously in Zhejiang University,specifically as follows(*Cr=Credits;*Sc=Score):
1.The percentage system:Above 60 is passing,100 is full mark.
2.Five degree grading:Excellent(A),Good(B),Fair(C),Passing(D),Failed(E).
3.Two degree grading:Passing(P),Failed(F).
4.Courses marked with * are courses transferred from other universities and their original records are kept.
5.Courses marked with "△" are repeat courses, and GPA is calculated on the highest grade.
6.Courses marked with "○" are not included in the GPA calculation.

Dean,Undergraduate School: 张光新 Date Issued:02/03/2022 00641

RECOGNITION INSTRUCTION: 1."ZHEJIANG UNIVERSITY" is microprinted on the lower right corner as a line, they can be seen under the magnifying glass. 2.The fluorescent school badge of ZHEJIANG University on the higher left corner will appear under the UV light. 3.The words "ZJU" on the center of the report will turn purple under the sunlight. 4.This style transcript has been formally in use since September 1,1999